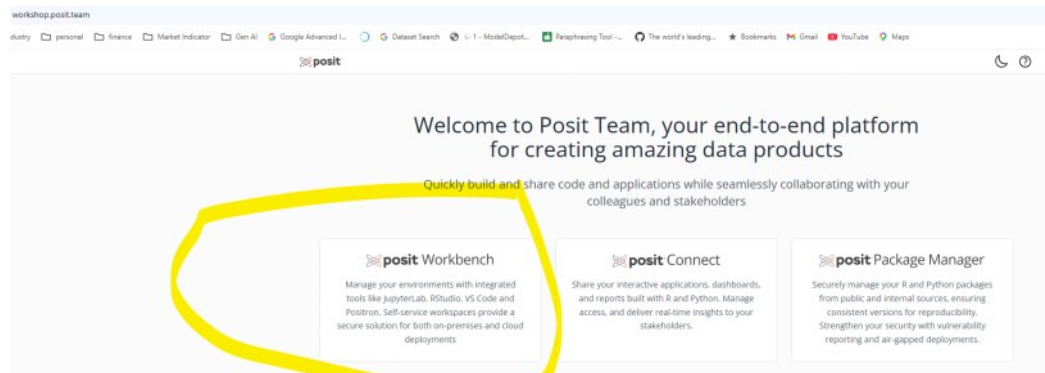


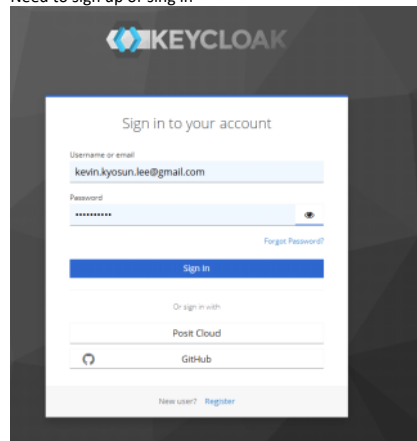
Starting Positron Pro Workshop

Saturday, September 19, 2026 5:09 PM

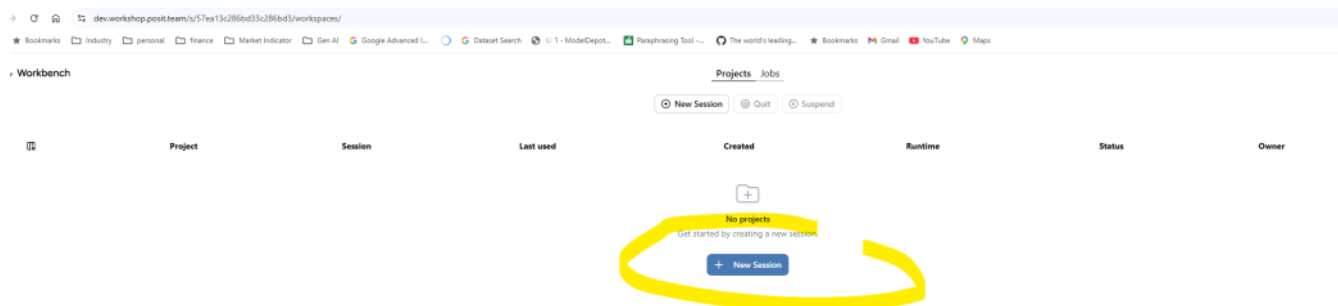
<https://workshop.posit.team/>



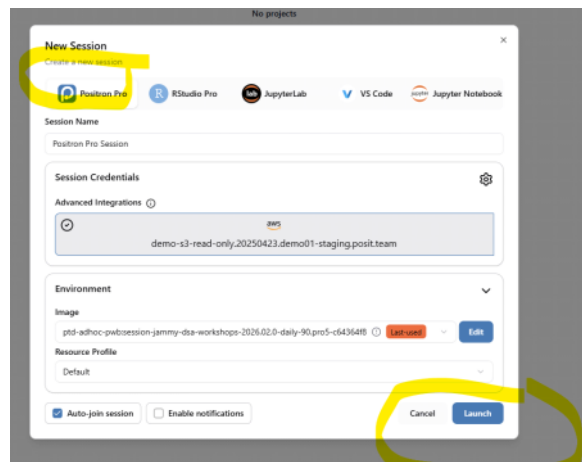
- Need to sign up or sing in



- Create the new session



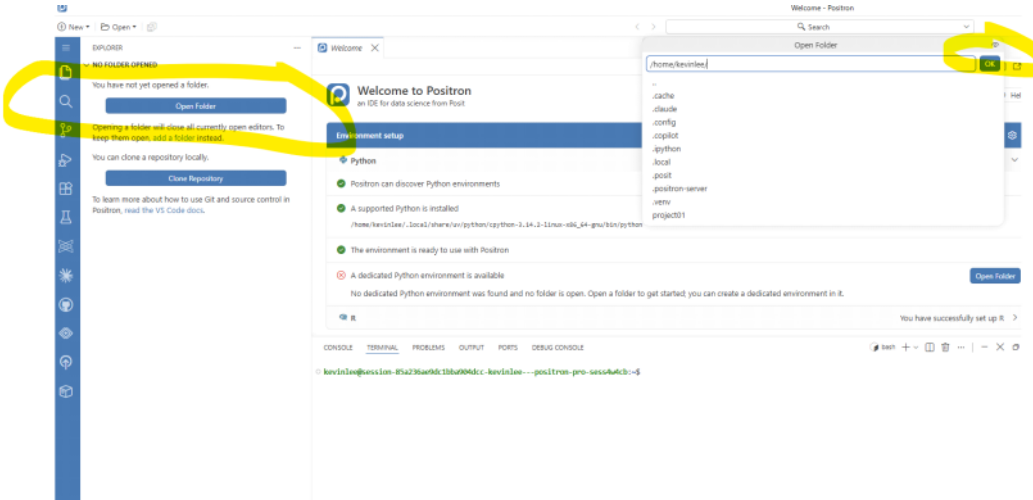
- Launch the Positron Pro New Session



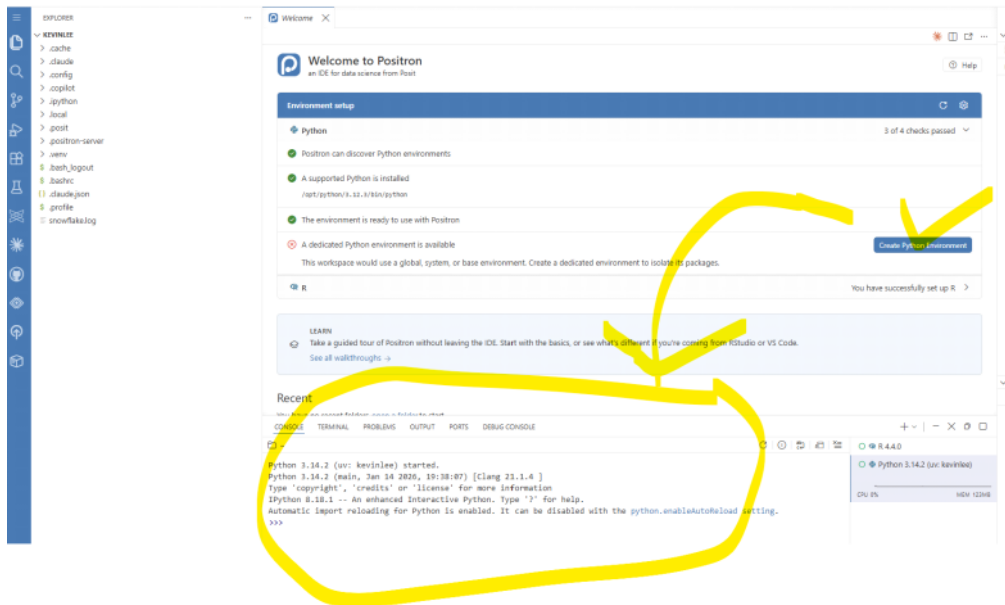
- Wait until it is launched

Project	Session	Last used	Created	Runtime	Status	Owner	Details
(no project)	Positron Pro Session	just now	just now		Pending		

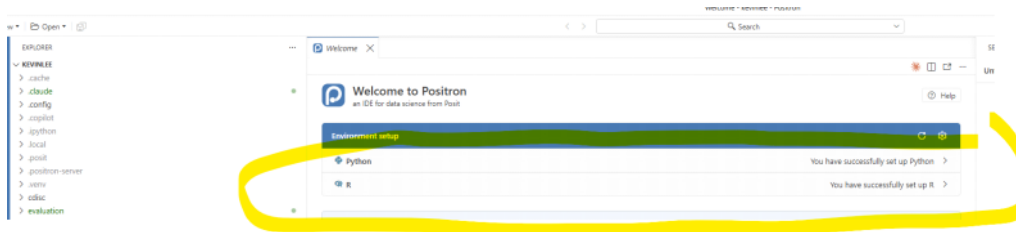
- Open the folder with your name



- Create Python Environment



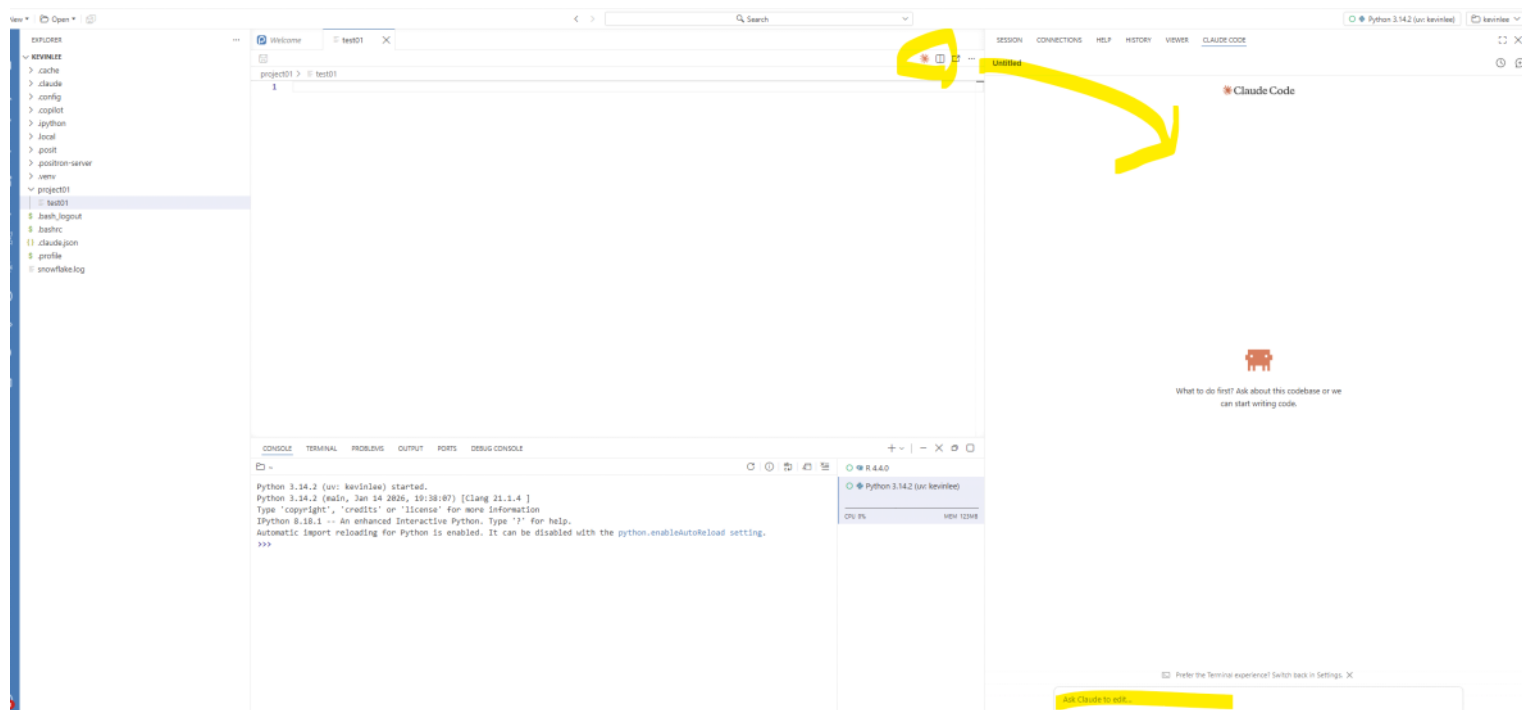
- Now, you have both python and R set up



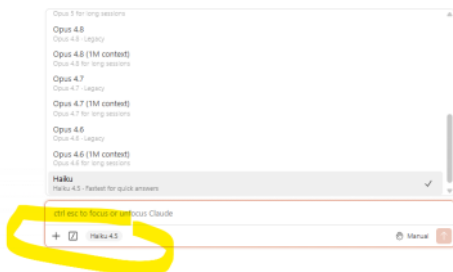
Starting Claude Code

Monday, September 28, 2026 3:25 PM

- Start Claude Code:



- Change the model to Haiku. Otherwise, your token might not be available as the end of course



Import cdisc data from github

Tuesday, September 29, 2026 5:05 PM

- In the terminal - git clone <https://github.com/cdisc-org/sdtm-adam-pilot-project.git>

```
kevinlee@session-85a236ae9dc1b09bb35a5-kevinlee---positron-pro-session5hkps:~/project02$ git clone https://github.com/cdisc-org/sdtm-adam-pilot-project.git
Cloning into 'sdtm-adam-pilot-project'...
remote: Enumerating objects: 224, done.
remote: Counting objects: 100% (224/224), done.
remote: Compressing objects: 100% (150/150), done.
remote: Total 224 (delta 64), reused 220 (delta 61), pack-reused 0 (from 0)
Receiving objects: 100% (224/224), 24.51 MiB | 2.08 MiB/s, done.
Resolving deltas: 100% (64/64), done.
Updating files: 100% (87/87), done.
```

```

└─ project02 / sdtm-adam-pilot-project
    └─ > reference-ranges
    └─ > updated-pilot-submission-package / 900172
        └─ > m1
        └─ > m5
            └─ > 53-clin-stud-rep
            └─ > datasets
                └─ > cdiscpilot01
                    └─ > analysis / adam
                        └─ > datasets
                            ├── {} adae.json
                            ├── ≡ adae.xpt
                            ├── {} adlbc.json
                            ├── ≡ adlbc.xpt
                            ├── {} adlbh.json
                            ├── ≡ adlbh.xpt
                            ├── {} adlbhy.json
                            ├── ≡ adlbhy.xpt
                            ├── {} adqsadas.json
                            ├── ≡ adqsadas.xpt
                            ├── {} adqscibc.json
                            ├── ≡ adqscibc.xpt
                            ├── {} adqsnpix.json
                            ├── ≡ adqsnpix.xpt
                            ├── {} adsl.json
                            ├── ≡ adsl.xpt
                            ├── {} adtte.json
                            └── ≡ adtte.xpt
```

```
- sample-
{} adtte.json
≡ adtte.xpt
{} advs.json
≡ advs.xpt
📄 dataguide.pdf
≡ define-v1-updated-html.xml
<> define.html
📄 define.pdf
📄 define.xml
> programs
▼ tabulations / sdtm
{} ae.json
≡ ae.xpt
📄 blankcrf.pdf
{} cm.json
≡ cm.xpt
≡ define-v1-updated-html.xml
<> define.html
📄 define.pdf
📄 define.xml
```

- Prompt in Claude:
download CDISC pilot SDTM and ADaM data from github in <https://github.com/cdisc-org/sdtm-adam-pilot-project/tree/master/updated-pilot-submission-package/900172/m5/datasets/cdiscpilot01>. It should be in cdisc folder , adam is adam subfolder and sdtm is in sdtm subfolder. Only xpt file.
-

Import skills in .claude/skills

Sunday, September 20, 2026 11:43 AM

- download skills of anthropic (<https://github.com/anthropics/skills/tree/main/skills>) in .claude/skills
- Prompt in Claude:
download skills from <https://github.com/anthropics/skills/tree/main/skills> to project01/skill-anthropic
- Copy skills to .claude/skills like below

The screenshot shows a code editor with two panels. The left panel displays a file explorer view of the .claude directory. The right panel shows the content of the SKILL.md file.

File Explorer (Left Panel):

- KEVINLEE
 - .cache
 - .claude
 - backups
 - .claude.json.backup.1788284590319
 - .claude.json.backup.1788462219860
 - .claude.json.backup.1789917830832
 - ide
 - 39104.lock
 - projects / -home-kevinlee
 - 6247802a-b99a-46c4-be58-997b1ad3dbd1
 - memory
 - 6247802a-b99a-46c4-be58-997b1ad3dbd1.jsonl
 - session-env
 - sessions
 - 5043.4e3d95268536f08d5cac563994ff122b6d67d74c9...
 - 5043.json
 - shell-snapshots
 - skills
 - academy-guide
 - algorithmic-art
 - brand-guidelines
 - canvas-design
 - claude-api
 - discernment-nudge
 - doc-coauthoring
 - docx
 - frontend-design
 - internal-comms
 - mcp-builder
 - pdf
 - pptx
 - skill-creator
 - agents
 - assets
 - eval-viewer
 - references
 - scripts
 - LICENSE.txt
 - SKILL.md
 - slack-gif-creator
 - theme-factory
 - web-artifacts-builder
 - webapp-testing
 - xlsx
 - .last-cleanup

SKILL.md Content (Right Panel):

```
1 ---
2 name: skill-creator
3 description: Create new skills, modify and improve existing skills, and measure skill performance
4   scratch, edit, or optimize an existing skill, run evals to test a skill, benchmark skill performance
5   description for better triggering accuracy.
6 ---
7 # Skill Creator
8 A skill for creating new skills and iteratively improving them.
9
10 At a high level, the process of creating a skill goes like this:
11
12 - Decide what you want the skill to do and roughly how it should do it
13 - Write a draft of the skill
14 - Create a few test prompts and run claude-with-access-to-the-skill on them
15 - Help the user evaluate the results both qualitatively and quantitatively
16 - While the runs happen in the background, draft some quantitative evals if there aren't any (if
17   already exist)
18 - Use the `eval-viewer/generate_review.py` script to show the user the results for them to look
19   at
```

Terminal Output (Bottom Panel):

```
.venv\kevinlee@session-85a236ae9dc1bba904dcc-kevinlee---positron-pro-sess4w4cb:~$ cd
bash: home: command not found
.venv\kevinlee@session-85a236ae9dc1bba904dcc-kevinlee---positron-pro-sess4w4cb:~$ ls
project01  snowflake.log
.venv\kevinlee@session-85a236ae9dc1bba904dcc-kevinlee---positron-pro-sess4w4cb:~$
```

- Check if you could use skill in claude

Import contents from RinPharma Workshop

Tuesday, September 29, 2026 5:51 PM

- `git clone --depth=1 https://github.com/kevinlee1004/2026\_RinPharma\_SKILL\_for\_AIAgent.git`

```
Updating files: 100% (18/18), done.
• .venv\kevinlee@session-85a236ae9dc1b09bb35a5-kevinlee---positron-pro-sess5hkps:~/project02$ git clone --depth=1 https://github.com/kevinlee1004/2026_RinPh
arma_SKILL_for_AIAgent.git
Cloning into '2026_RinPharma_SKILL_for_AIAgent'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 2.20 MiB | 3.62 MiB/s, done.
```

- Move to skills in .clause/skills so that we could use it.

```
▼ project02
  ▼ 2026_RinPharma_SKILL_for_AIAgent
    ▼ skills
      > KL-disp-analysis
      > KL-patient-profile
      ⓘ README.md
      📄 Skills_AIAgents_v01.pdf
```

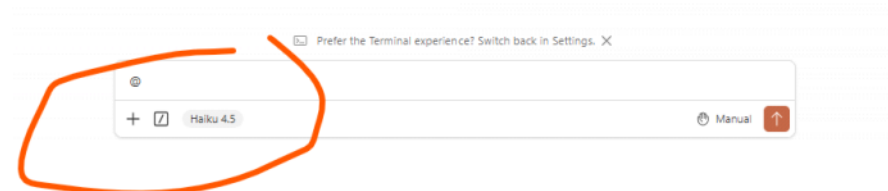
Use case claude code

Tuesday, September 29, 2026 1:00 PM

- / to find tool and skill
- @ to find the document
- New chat / previous chat
- Upload the document



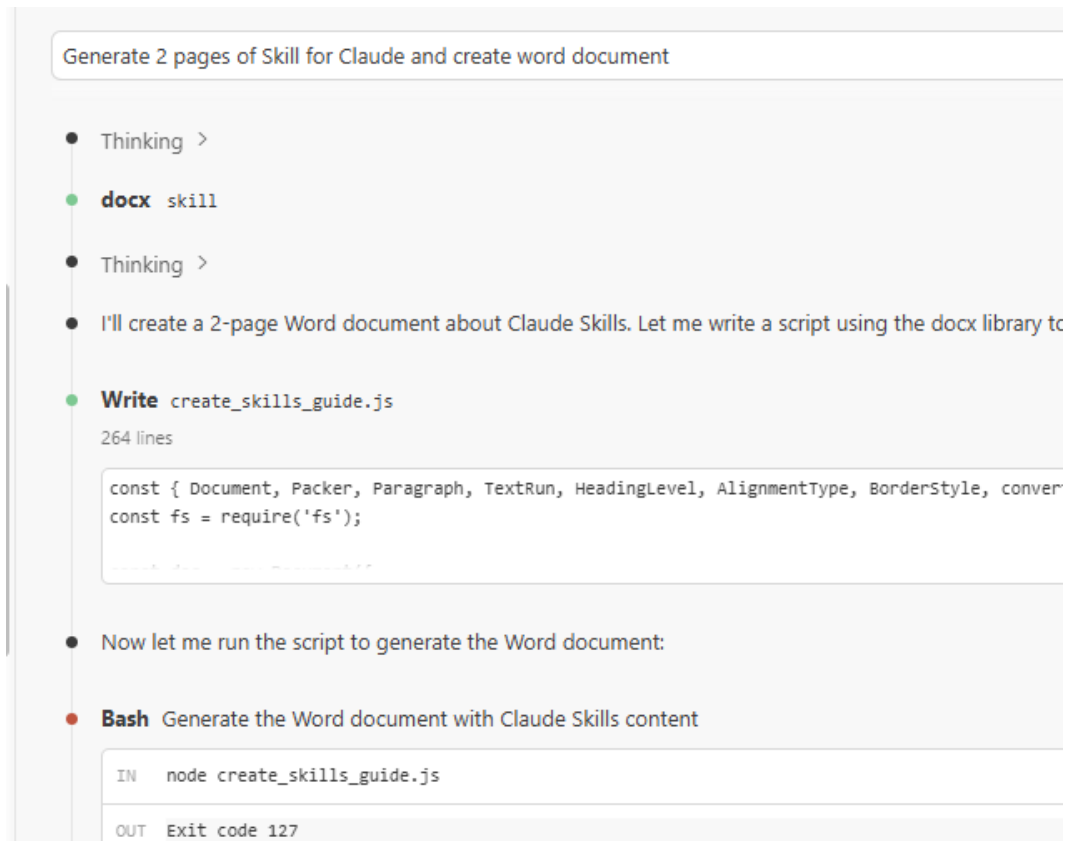
Use Claude Code in the terminal to configure
MCP servers. They'll work here, too!



Use case 1 - use the existing skills

Tuesday, September 29, 2026 12:42 PM

- Prompt:
Generate 2 pages of Skill for Claude and create word document



```
Generate 2 pages of Skill for Claude and create word document

● Thinking >
● docx skill
● Thinking >
● I'll create a 2-page Word document about Claude Skills. Let me write a script using the docx library to

● Write create_skills_guide.js
  264 lines

  const { Document, Packer, Paragraph, TextRun, HeadingLevel, AlignmentType, BorderStyle, conver
  const fs = require('fs');

  ...

● Now let me run the script to generate the Word document:

● Bash Generate the Word document with Claude Skills content

  IN  node create_skills_guide.js

  OUT Exit code 127
```

-  **Document created successfully!**
I've generated a 2-page Word document titled **Claude_Skills_Guide.docx** located at `/home/kevinlee/Claude_Skills_Guide.docx` (38KB).
The document covers:
Page 1:
 - What Are Skills? — Introduction to the concept and purpose
 - When to Use Skills — 5 key scenarios for skill application
 - Popular Built-in Skills — Overview of dataviz, pdf, docx, code-review, and run skills**Page 2:**
 - How Skills Work — Mechanism and capabilities of skills
 - Creating Custom Skills — Structure and components for building custom skills
 - Best Practices — 5 key principles for skill development
 - Conclusion — Summary of skills' value and how to get started

The document is professionally formatted with headings, bullet points, and proper spacing for readability.

- Prompt: Create power point slides

create the power point slides with 2 pages.

● Thinking >

● **pptx** skill

● Thinking >

● I'll create a professional 2-slide PowerPoint presentation about Claude

● **Write** create_skills_pptx.js

199 lines

```
const PptxGenJS = require("pptxgenjs");

const pres = new PptxGenJS();

// ... Thinking ...
```

● Now let me run the script to generate the PowerPoint:

●  **PowerPoint presentation created successfully!**

I've generated a professional 2-slide PowerPoint presentation titled **Claude_Skills_Guide.pptx** (30KB) with modern design:

Slide 1: Title Slide

- Navy blue background with elegant typography
- Bold title: "Claude Skills"
- Subtitle: "Extending Claude Code with Custom Capabilities"
- Accent mint-green line for visual polish

Slide 2: Content Slide

- Header with deep blue background
- Left column with key capabilities (5 bullet points)
- Right column with popular built-in skills highlighted in a light gray box:
 - dataviz, pdf, docx, code-review, run
- Bottom call-to-action banner with mint accent
- Professional color palette (navy, deep blue, mint green)

Both files are now ready:

-  **Claude_Skills_Guide.docx** (38KB) - Word document with 2 pages
-  **Claude_Skills_Guide.pptx** (30KB) - PowerPoint with 2 slides

Use case 2 - Data analysis

Tuesday, September 29, 2026 12:12 PM

- Ad hoc analysis
 - Prompt:
find how many patients have AE from @cdisc/sdtm/ae.xpt

</> ae.xpt

find how many patients have AE from @cdisc/sdtm/ae.xpt

● Thinking >

● I'll read the SDTM AE dataset and count the unique patients with adverse events.

● **Bash**

```
IN python3 << 'EOF'
import pyreadstat
```

```
OUT Exit code 1
Traceback (most recent call last):
  File "<stdin>", line 4, in <module>
```

● Let me fix that—the correct function name is read_xport:

● **Bash**

```
IN python3 << 'EOF'
import pyreadstat
```

```
OUT Total AE records: 1191
Unique patients with AE: 225
```

● **225 patients** have adverse events in the ae.xpt dataset. There are 1,191 total AE records across these patients.

- Create excel output based on this analysis, and save it in project01 folder. name the file starting with the current date and time in format of yyyy-mm-dd-hr-mm and input file name.

• Thinking >

• Bash

```
IN  ls -la /home/kevinlee/project01/ 2>/dev/null || echo "project01 folder does not exist, will create it"
```

```
OUT total 10
drwx----- 2 kevinlee kevinlee 3 Sep 20 16:38 .
drwx----- 15 kevinlee kevinlee 29 Sep 28 19:24 ..
```

• Now I'll create an Excel file with the AE analysis and save it with the proper naming format:

• Bash

```
IN  python3 << 'EOF'
import pyreadstat
import pandas as pd
```

```
OUT Exit code 1
Traceback (most recent call last):
  File "<stdin>", line 24, in <module>
```

• Thinking >

• Bash

```
IN  pip install openpyxl -q && python3 << 'EOF'
import pyreadstat
import pandas as pd
```

```
OUT ✓ Excel file created: 2026-09-29-16-20_ae.xlsx
✓ Saved to: /home/kevinlee/project01/2026-09-29-16-20_ae.xlsx
```

• Done! Created **2026-09-29-16-20_ae.xlsx** in `/home/kevinlee/project01/` with 5 sheets:

- **Summary** — key statistics (225 patients, 1,191 total AE records)
- **AE by SOC** — adverse events grouped by System Organ Class
- **Top AE Terms** — 30 most common adverse event terms
- **Patient AE Counts** — AE frequency per patient
- **All Records** — complete detailed dataset

- Create the skill

Use Case 3 - Vibe coding to SKILLS

Sunday, September 20, 2026 5:13 PM

- Vibe coding:

Write R script to read `@cdisc/sdm/ds.xpt` to provide patient status in .md file. and run R script to create .md file.

The screenshot shows a file explorer on the left with a directory structure including .cache, .cdisc, .config, .copilot, .local, .post, .postion-server, .new, .cdisc, .project01, .bash_history, .bash_logout, .bashrc, .cdisc.json, .gitignore, .profile, .make_profile_path, .patient_status.xpt, .patient_status.md, .profile_01-701-1015.md, .profile_01-701-1015.pdf, .profile_01-701-1015.pdf, .README.md, and .sourcefile.log. The main preview window displays the content of `patient_status.md`, which includes a status summary table and a table of reasons for discontinuation.

Status	N	Percent
Completed	110	35.9%
Discontinued	144	47.1%
Screen Failure	52	17.0%
Total	306	100.0%

Reason (DSDECOD)	N	% of Discontinued
ADVERSE EVENT	92	63.9%
WITHDRAWAL BY SUBJECT	27	18.8%
STUDY TERMINATED BY SPONSOR	7	4.9%
PROTOCOL VIOLATION	6	4.2%
LACK OF EFFICACY	4	2.8%
DEATH	3	2.1%
PHYSICIAN DECISION	3	2.1%
LOST TO FOLLOW-UP	2	1.4%

- Prompt:

Convert this to skill.

create the skill for above action using skill-creator skill.

Do not include patient level report in output.

structure should be as follow.

input:

- language choice R or Python
- input data location
- output scripts and file location

output: .md file

process:

- 1.
- 2.
- 3.

Notes:

The screenshot shows a skill preview window for a skill named "Disposition analysis from SDTM DS". The skill description states: "Produces a study-level markdown report of subject disposition — how many subjects completed, how many discontinued and why, how many never got past screening — from an SDTM DS dataset." It also mentions that two bundled scripts do the work, one per language, and they are written to produce byte-identical reports. The input/output details are as follows:

Input	Required	Notes
Language	No — but ask if unclear	R or Python. Ask when the user has not said and there is no signal in the conversation: sponsors are usually standardized on one, so guessing wastes their time. If they clearly don't care, use R — R is the lighter dependency.
Input data path	Yes	Either ds.xpt itself or a study directory. Both scripts walk a directory tree for ds.xpt (case-insensitive), so a study root with sdta/ underneath is fine. If missing, ask and stop — nothing downstream works without it.
Output script path	No	Where to leave the analysis script. Defaults to the working directory. The script is a deliverable, not a temp file: the user re-runs it when the data refreshes, so new is not of the skill rather than modify it in place.

- Schedule skill

The screenshot shows a skill preview window for a skill named "scheduled_tasks.json". The skill description states: "Produce a JSON file of tasks." The input/output details are as follows:

Input	Required	Notes
Language	No — but ask if unclear	R or Python. Ask when the user has not said and there is no signal in the conversation: sponsors are usually standardized on one, so guessing wastes their time. If they clearly don't care, use R — R is the lighter dependency.
Input data path	Yes	Either ds.xpt itself or a study directory. Both scripts walk a directory tree for ds.xpt (case-insensitive), so a study root with sdta/ underneath is fine. If missing, ask and stop — nothing downstream works without it.
Output script path	No	Where to leave the analysis script. Defaults to the working directory. The script is a deliverable, not a temp file: the user re-runs it when the data refreshes, so new is not of the skill rather than modify it in place.

Use case 4 - building skill

Sunday, September 20, 2026 11:54 AM

- Call the skill-creator to start building skill



- Prompt to create skill generator.

I want to create the skill that create the patient profile. Input should be the location of data either in SDTM and ADaM folder & patient number & data that you want to use - demo, disposition, AE, Lab, vital sign and so on. output is the .md file to have patient information.

Its format should be like

Input:

Output:

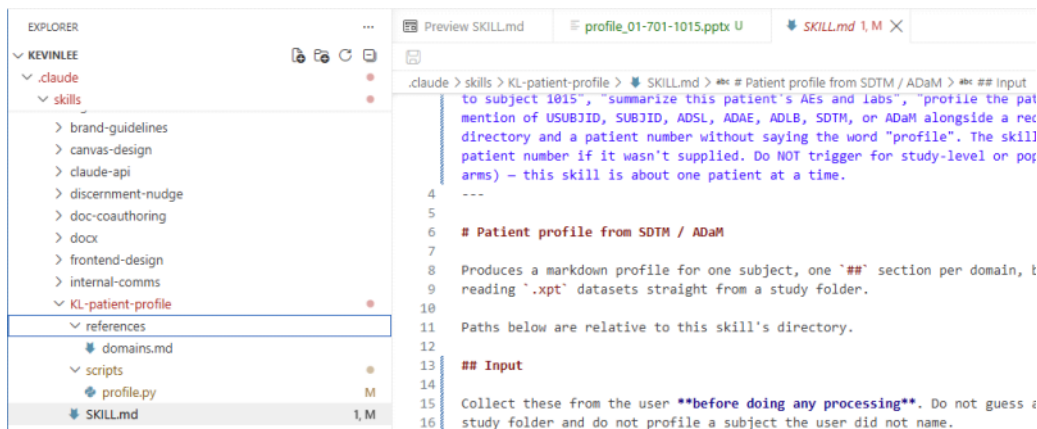
process is

step 1. identify the input. If not given, ask the user.

step 2. identify patient number. If not given, read dm or adsl, provide the list of the patients

step 3. write the python codes to read patient info based given data. The data is .xpt file format.

step 4. create .md file based on the results from step 3.



- Create the patient profile using SDTM DM, AE, LAB, DS on the patient of 01-701-1015.

The screenshot shows a code editor interface. On the left, the 'EXPLORER' panel displays a project structure under 'KEVINLEE'. The 'project01' folder is expanded, showing 'test01', '.bash_history', '.bash_logout', '.bashrc', '.claude.json', '.gitignore', '.profile', 'make_profile_pptx.py', 'profile_01-701-1015.md' (U), 'profile_01-701-1015.pdf' (U), 'profile_01-701-1015.pptx' (U), 'README.md', and 'snowflake.log'. The main editor shows a preview of the 'profile_01-701-1015.md' file. The preview includes a title 'Patient Profile: 01-701-1015', a generation date 'Generated 2026-09-20 from ae.xpt, dm.', a 'Demographics' section with a table of patient information, and a 'Disposition' section with a table of treatment events.

Field	Value
Study	CDISCILOT01
Site	701
Age	63 YEARS
Sex	F
Race	WHITE
Ethnicity	HISPANIC OR LATINO
Country	USA
Planned arm	Placebo
Actual arm	Placebo
First dose	2014-01-02
Last dose	2014-07-02

Day	Date	Category
-----	------	----------

Use case 5 - how to use skill

Sunday, September 20, 2026 12:01 PM

- Prompt:
use [@profile_01-701-1015.md](#) to create pdf file.

The screenshot shows the Claude Code interface with a terminal window titled "Profile 01-701-1015 PDF generation". The terminal contains the following content:

```
/clear
```

use: [@profile_01-701-1015.md](#) to create pdf file.

- Thinking
- I'll use the PDF skill for this.
- pdf skill
- Thinking
- **Bash** Check available PDF tooling

```
IN  which pandoc weasyprint wkhtmltopdf xelatex pdflatex 2>/dev/null; python3 -c "import reportlab, markdown; print('reportlab+markdow
```

```
OUT /usr/bin/pandoc
    /usr/local/bin/xelatex
    /usr/local/bin/pdflatex
```

- Thinking
- **Bash** Convert markdown to PDF with pandoc

```
IN  pandoc profile_01-701-1015.md -o profile_01-701-1015.pdf --pdf-engine=xelatex -V geometry:landscape -V geometry:margin=0.6in -V fo
```

```
OUT -FW-----, 1 kevinlee kevinlee 23916 Sep 20 20:39 profile_01-701-1015.pdf
```

- Thinking
- **Bash** Render PDF pages to images

```
IN  cd /tmp && pdftoppm -png -r 88 /home/kevinlee/profile_01-701-1015.pdf pg && ls pg*
```

```
OUT The user doesn't want to proceed with this tool use. The tool use was rejected (eg. if it was a file edit, the new_string was NOT
```

Tool interrupted

EXPLORER

KEVINLEE

> .cache

> .claude

> .config

> .copilot

> .ipython

> .local

> .posit

> .positron-server

> .venv

> cdisc

project01

test01

.bash_history

.bash_logout

.bashrc

.claude.json

.gitignore

.profile

make_profile_pptx.py

profile_01-701-1015.md

profile_01-701-1015.pdf

profile_01-701-1015.pptx

README.md

snowflake.log

Preview profile_01-701-1015.md

profile_01-701-1015.pdf U X

profile_01-701-1015.pdf

1 of 3

Pat

Patient Profile: 01-701-1015

Generated 2026-09-20 from ae.xpt, dm.xpt, ds.xpt, lb.xpt

Demographics

Disposition

Adverse Events

Day	Date	Category
182	2014-07-02	DISPOSITIO
182	2014-07-02	OTHER EV

CONSOLE

TERMINAL

PROBLEMS

OUTPUT

PORTS

DEBUG CONSOLE

COMMENTS

Other skills

Sunday, September 20, 2026 5:41 PM

- R consortium in Pharma : <https://github.com/RConsortium/pharma-skills/tree/main>
- Anthropic skills : <https://github.com/anthropics/skills/tree/main/skills>
- Popular Agent skills : <https://github.com/topics/agent-skills>